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ITEM PACKAGING MACHINE

Abstract

An item packaging machine is configured to fulfill a plurality of orders that each include one or more items. The item packaging machine includes an introduction assembly configured to index each item in a retaining notch that is advanced by a conveyor over a base. The conveyor advances each item to a label applicator assembly, where each item is verified by one or more cameras directed to at least one side of each item and a label is applied to each item. The conveyor advances each labeled item to a packaging assembly, where each item is verified and advanced into a package that includes order identification information applied thereon and that is configured to package the order that includes the item.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Patent Application No. 63/559,676 filed Feb. 29, 2024, the disclosure of which is hereby incorporated herein in its entirety by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] This invention relates generally to packaging machines, and more specifically to an assembly and method for verifying and packaging pharmaceutical items.

BACKGROUND

[0003] Various solutions exist for labeling and packaging items in order fulfillment operations. In traditional prescription fulfillment operations, a packaging clerk and/or a pharmacist labels a bag, verifies a prescription, labels the prescription, and packs the prescription and associated documentation in the bag for shipping.

SUMMARY

[0004] This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the detailed description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Other aspects and advantages of the invention will be apparent from the following detailed description of the embodiments and the accompanying drawing figures.

[0005] The embodiments described herein relate to an item packaging machine configured to fulfill a plurality of orders. Each of the plurality of orders includes one or more items, each having an item identifying indicia thereon. A processing system accesses order information associated with each of the plurality of orders and includes order identifying indicia for each of the plurality of orders. A conveyor is configured to advance the one or more items through the item packaging machine, and the conveyor forms a conveyor run on which each of the one or more items are advanced. An introduction assembly is configured to associate an indexed position of each of the one or more items positioned on the conveyor run with item identifying information for the item at the respective indexed position. A label applicator is positioned proximate to a labeling station on the conveyor run, and a least one labeling verification camera is positioned proximate the label applicator and in communication with the processing system. Upon each of the items being advanced to the labeling station, the processing system causes the at least one labeling verification camera to capture the item identifying indicia on the item at the labeling station, and if the item identifying indicia captured from the item at the labeling station corresponds to the item identifying indicia associated in the processing system with the item expected at the labeling station based upon the indexed position of the item in the processing system, the processing system causes the label applicator to apply to the item a label with customer identifying indicia associated with the item printed thereon. Thereafter, the processing system causes the at least one labeling verification camera to capture at least one image of the item wherein the at least one image includes the item identifying indicia and at least a portion of the label including the customer identifying indicia thereon. A packaging verification camera is positioned proximate to a packaging verification station on the conveyor run proximate a packaging assembly operable to advance the item from the packaging verification station into a package, and the packaging verification camera and the packaging assembly are in communication with the processing system. The processing system causes the packaging verification camera to capture the item identifying indicia on the item or the

customer identifying indicia on the label at the packaging verification station, and if the item identifying indicia or the customer identifying indicia captured from the item at the packaging verification station corresponds to the item identifying indicia associated in the processing system with the item expected at the packaging verification station based upon the indexed position of the item in the processing system, the processing system causes the packaging assembly to advance the item into a package and to apply the order identifying information for the order to the package.

[0006] In an embodiment, the conveyor further includes a plurality of V-shaped blocks, and the indexed position of the item corresponds to one of the plurality of V-shaped blocks.

[0007] In an embodiment, the introduction assembly includes an introduction camera that is communicatively coupled to the processing system. The processing system causes the introduction camera to capture an image of the item identifying indicia on each of the items as the item is advanced into one of the indexed positions on the conveyor run to associate the item identifying indicia with the indexed position and to verify the item identifying indicia corresponds to the item identifying indicia of one of the one or more items in the order.

[0008] In an embodiment, a display is communicatively coupled to the processing system. Upon capturing the image of the item identifying indicia of the item by the introduction camera, the display provides a prompt indicating an orientation of the item identifying indicia on the item relative to the conveyor when positioned on the conveyor run.

[0009] In an embodiment, the at least one labeling verification camera includes a set of six labeling verification cameras, and each of the six labeling verification cameras is positioned relative to the labeling station to image a different side of the item.

[0010] In an embodiment, the label applicator is mounted for movement along at least two axes, and two of the axes extend in a plane extending generally perpendicular to the conveyor run across the labeling station.

[0011] In an embodiment, the conveyor run across the labeling station includes an item receiving platform advanceable along an axis perpendicular to the conveyor run across the labeling station for raising or lowering the item receiving platform relative to the conveyor run at the labeling station. The at least one labeling verification camera includes a set of six labeling verification cameras wherein each of the six labeling verification cameras is positioned relative to the labeling station to image a different side of the item on the item receiving platform.

[0012] In an embodiment, the order information accessed by the processing system includes a print file including the customer identifying indicia and patient-specific information for each of the one or more items in the order to be printed on the label, a location for applying the label containing the patient-specific information and the customer identifying indicia relative to a side of the item positioned on the item receiving platform, and a position of the item receiving platform relative to the conveyor run to position the item so that one of the six labeling verification cameras can capture an image of the side of the item on which the item identifying information is located.

[0013] In an embodiment, the packaging assembly further includes a document verification camera positioned proximate to the packaging assembly and in communication with the processing system. The processing system causes the document verification camera to capture an image of a document verification indicia applied to one or more patient information documents associated with at least one of the items in the order and to compare the document verification indicia captured by the document verification camera to the document verification indicia associated with each of one or more patient information documents associated with the order in the processing system prior to insertion of the one or more patient information documents into the package for each order.

[0014] The embodiments described herein relate to a method of fulfilling a plurality of orders that each include one or more items for processing in an item packaging machine. The method includes retrieving from a database, order information regarding an order including item identification information corresponding to each of the one or more items contained in the order and customer information of a customer associated with the order. The method further includes imaging an item

identification indicia of each of the one or more items in succession, wherein each item identification indicia is associated with the item identification information for a respective one of the items in the order. For each of the one or more items whose item identification indicia is imaged, the method includes prompting placement of the item onto a conveyor in a specified orientation. The method further includes positioning the item of the order onto the conveyor with the item identification indicia on the item in the specified orientation and at a specified position on the conveyor. Still further, the method includes conveying the item to a verification position on the conveyor. At the verification position, the method includes imaging the item identification indicia on the item and comparing the item identification indicia to the item identification information in the order information retrieved from the database, and if the item identification indicia on the item corresponds to the item identification information in the order, applying customer information associated with the item of the order to the item. After applying the customer information to the item and with the item at the verification position, the method includes capturing verification images that include an image of the item identification indicia on the item and an image of the customer information applied to the item.

[0015] In an embodiment, the item identification indicia and the customer information are imaged simultaneously.

[0016] In an embodiment, the verification position is on a vertically moving platform.

[0017] In an embodiment, the method includes performing a pharmaceutical verification of the item. The pharmaceutical verification includes: imaging a selected side of the item prior to applying the customer information to the item; compiling into a file the verification images, the image of the selected side of the item prior to applying the customer information thereto, and a pre-existing image from the item identification information from the database that corresponds to the selected side of the item; comparing the images in the file with at least the item identification information retrieved from the database; and confirming the item is the item in the order or sending the item to be reviewed.

[0018] In an embodiment, the order information retrieved from the database includes address information for the customer. The method further includes conveying the item to a packaging verification position. At the packaging verification position, the method includes imaging the customer information on the item and comparing the customer information to the item identification information in the order information retrieved from the database and if the customer information on the item at the packaging verification position corresponds to the item identification information in the order for the item expected at the packaging verification position, the method includes advancing the item into a package. Still further, the method includes applying the address information for the customer on the package into which the item is to be placed.

[0019] In an embodiment, the order information retrieved from the database includes order identification indicia, and the method further includes applying the order identification indicia on the package into which the item is to be placed and imaging the order identification indicia.

[0020] The embodiments described herein relate to a method of fulfilling a plurality of orders that each include one or more items for processing in an item packaging machine. The method includes receiving an order container at an item introduction assembly containing one or more items corresponding to an order and imaging a container identification indicia of the order container. The method further includes retrieving from a database order information associated with the container identification indicia imaged at the item introduction assembly. The order information corresponds to the order, including the one or more items contained in the order container imaged at the item introduction assembly. The method further includes imaging an item identification indicia of each of the one or more items retrieved in succession from the order container. For each of the one or more items whose item identification indicia is imaged, the method includes prompting placement of the item onto a conveyor in a specified orientation. The method further includes placing the item onto the conveyor in a specified position in the specified orientation. The method further includes

verifying a presence of the item on the conveyor. Still further, the method includes indexing the specified position of the item on the conveyor. The method further includes incrementally advancing the conveyor with the item in the specified position towards a labeling position. Upon advancement of the item to the labeling position, the method includes verifying the item and the specified orientation of the item by imaging the item identification indicia on the item and comparing the item identification indicia to the order information and to the item expected in the specified position. The method further includes applying patient-specific information to the item in the labeling position. The method further includes verifying the item by imaging, in the labeling position, the patient-specific information applied to the item and comparing the patient-specific information to the order information. Still further, the method includes incrementally advancing the conveyor with the item in the specified position towards a packaging assembly. At the packaging assembly, the method includes verifying the item by imaging the patient-specific information applied to the item and comparing the patient-specific information to the order information and to the item expected in the specified position. The method further includes applying order identification information to a package, and verifying the package by imaging the order identification information applied to the package and comparing the order identification information with the order information. Still further, the method includes advancing the one or more items of the order into the package. Still further, the method includes sealing the package. [0021] In an embodiment, the conveyor includes a plurality of V-shaped blocks, and the specified position of the item corresponds to one of the plurality of V-shaped blocks.

[0022] In an embodiment, the method further includes verifying the item by imaging the item identification indicia of the item simultaneously with imaging of the patient-specific information applied to the item and comparing the item identification indicia to the order information.

[0023] In an embodiment, the method further includes verifying the item further includes imaging at least one surface of the item prior to applying the patient-specific information thereto. Still further, the method includes compiling a set of images that includes: the image of the patient-specific information applied to the item, the image of the item identification indicia of the item taken simultaneously with the image of the patient-specific information applied to the item, the image of the at least one surface of the item prior to applying the patient-specific information thereto, and at least one pre-existing image that corresponds to the image of the at least one surface of item. Still further, the method includes comparing the set of images with the order information. Yet still further, the method includes operating an input device to confirm the labeled item is the item in the order or send the item for review.

[0024] In an embodiment, the labeling position is on an item receiving platform that is vertically positionable in an upper position and a lowered position, and imaging in the labeling position is effected by a set of one or more cameras that are vertically positionable with the item receiving platform.

[0025] In an embodiment, the method further includes printing one or more patient information documents with one or more printers. Still further, the method includes imaging each of the one or more patient information documents to verify that the one or more patient information documents is associated with the one or more items of the order. The method further includes advancing the one or more patient information documents into the package prior to sealing the package.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] Embodiments of the invention are described in detail below with reference to the attached drawing figures.

[0027] FIGS. **1a** and **1b** show front perspective views of embodiments of an item packaging

machine assembly (the “IPM assembly”).

[0028] FIG. 2 shows a front perspective of an embodiment of a tote delivery conveyor as shown in FIGS. 1a and 1b, with a tote positioned thereon, for delivery of totes to the IPM assembly, and a chute and a tote recovery conveyor for advancing totes away from the item packaging machine.

[0029] FIG. 3 shows a rear perspective view of an embodiment of the IPM assembly, as shown in FIG. 1a.

[0030] FIG. 4 shows a perspective view of an embodiment of a plurality of V-shaped blocks consecutively positioned on a base to index items on a conveyor run of the IPM assembly, as shown in FIGS. 1a and 1b.

[0031] FIGS. 5a-5g show views of an item with a barcode positioned relative to a retaining notch of the V-shaped block of FIG. 4 with arrows indicating a direction that a side of the item with the barcode is facing, similar to an operator visual aid that might be shown on a monitor to indicate the direction that the barcode on the item is to face when the item is introduced into the IPM assembly.

[0032] FIGS. 6a and 6b show front elevation views of embodiments of an item induction assembly of the IPM assembly shown in FIGS. 1a and 1b, respectively, including an optical imaging device, a processing system with a monitor, and a sensor for sensing a presence of each item introduced into the item induction assembly.

[0033] FIG. 7 shows a front and side perspective view of an embodiment of a label applicator assembly of the IPM assembly shown in FIGS. 1a and 1b, including a lowerator platform in an upper position and with a label applicator removed.

[0034] FIG. 8 shows a front and side perspective view of the label applicator assembly of the IPM assembly shown in FIGS. 1a and 1b, including the lowerator platform in a lowered position and with the label applicator removed.

[0035] FIG. 9 shows a perspective view from above of a set of cameras positioned relative to an item retained in the retaining notch of the V-shaped block and on the lowerator platform of an embodiment of the label applicator assembly shown in FIGS. 1a and b, and a tamp of the label applicator is positioned proximate to the item.

[0036] FIGS. 10a and 10b show perspective views of an embodiment of the superstructure shown in FIG. 1a and an embodiment of the mini-structure shown in FIG. 1b, respectively, moveably supporting the label applicator therein.

[0037] FIGS. 11a-11c show schematic views of a tamp engaging items of different shapes and sizes and positioned in the retaining notch of the V-shaped block for application of the label to the item.

[0038] FIG. 11d shows a perspective view of the items shown in FIGS. 11a-11c with labels applied thereto.

[0039] FIG. 12 shows a front perspective view of an embodiment of a packaging assembly as shown in FIG. 1a, including an imaging device for verifying the item, a drop gate, and a chute through which an item or items are advanced into a package.

[0040] FIGS. 13a and 13b show front perspective views of embodiments of a packaging assembly shown in FIGS. 1a and 1b, respectively.

[0041] FIGS. 14a-14c are a flow diagrams of an embodiment of an item packaging machine process that is part of an order fulfillment system and includes steps of inducting, labeling, and packaging one or more items using the IPM assembly shown in FIGS. 1a and 1b.

[0042] FIG. 15 shows a schematic view of a predetermined set of images of the item to verify the item in a pharmaceutical verification process.

[0043] The drawing figures do not limit the invention to the specific embodiments disclosed and described herein. The drawings are not necessarily to scale, emphasis instead being placed upon clearly illustrating the principles of the invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0044] The following detailed description references the accompanying drawings that illustrate specific embodiments in which the invention can be practiced. The embodiments are intended to

describe aspects of the invention in sufficient detail to enable those skilled in the art to practice the invention. Other embodiments can be utilized, and changes can be made without departing from the scope of the invention. The following detailed description is, therefore, not to be taken in a limiting sense. The scope of the invention is defined only by the appended claims, along with the full scope of the equivalents to which such claims are entitled.

[0045] In this description, references to “one embodiment,” “an embodiment,” or “embodiments” mean that the feature or features being referred to are included in at least one embodiment of the technology. Separate references to “one embodiment,” “an embodiment,” or “embodiments” in this description do not necessarily refer to the same embodiment and are also not mutually exclusive unless so stated and/or except as will be readily apparent to those skilled in the art from the description. For example, a feature, structure, act, etc. described in one embodiment may also be included in other embodiments but is not necessarily included. Thus, the technology can include a variety of combinations and/or integrations of the embodiments described herein.

[0046] The term capture, and variants thereof, may be used herein synonymously with variants of the terms image, read, and scan. The terms comprise translating indicia or light into an image or numbers or other information, and may further comprise accessing information associated with the indicia. The term may also comprise recording the translated indicia or light onto a digital medium or physical medium.

[0047] Embodiments of item packaging machines are shown in FIGS. **1a** and **1b**, which may each be referred to as an IPM assembly **10** and function similarly. The IPM assembly **10** described herein may be part of an order fulfillment system that is used to fulfill orders, such as, for example, orders for prescription and/or non-prescription pharmaceutical products or other products.

[0048] An order and corresponding order information is sent for each order to a database that is communicatively coupled to the order fulfillment system, which includes the IPM assembly **10**. Each order corresponds to or is married to an item container or tote **11** that is configured to contain one or more items **12** of the order, and the tote **11** may contain a multi-item order, a single-item order, or multiple single-item orders. The order information is retrieved from the database and includes item identification information corresponding to each of the one or more items **12** contained in the order. Each item **12** in the order is picked, imaged, and placed into the tote **11**, which may be referred to as picked-and-placed, by a process that includes imaging a tote identifying indicia **13** on the tote **11** to verify or confirm that the item **12** that is picked-and-placed into the tote **11** corresponds to the item identification information of one of the one or more items **12** in the order. A plurality of totes **11**, each containing one or more orders, is delivered to the IPM assembly **10** and singulated for fulfillment of each order.

[0049] As shown in FIG. **2**, each tote **11** includes the tote identifying indicia or identification tag **13**, such as a radio frequency identification (“RFID”), a quick response (“QR”) code, or a barcode, that corresponds to the one or more orders contained in the tote **11**. Prior to introducing and processing the item or items **12** of the one or more orders in the tote **11** into the IPM assembly **10**, the identification tag **13** is read by an identification tag reader or identification imager **14**. The identification tag reader captures or reads a bar code or other indicia on the identification tag **13** on the tote **11** which is communicated to a processing system or director **15** of the IPM assembly **10**. The processing system **15** is configured to communicate with components or assemblies of the IPM assembly **10** and/or with a processing system of the order fulfillment system. The processing system **15** retrieves the order information from the database for the order or orders that correspond to the bar code or other indicia on the identification tag **13** of the tote **11**, including item identification information corresponding to each of the one or more items **12** contained in each order, packaging information, and labeling information that includes customer information for a customer or patient that is associated with each item of the order.

[0050] Each tote **11** is singulated and delivered to the IPM assembly **10** via a tote delivery conveyor **20**. Singulation may be by a manual process or automated process known in the art. After

all the or one or more items **12** of the one or more orders are removed from the tote **11**, the tote **11** is advanced to a tote recovery conveyor **22** via a transition member or chute **24**. In an embodiment, when each of the items **12** of the one or more orders in the tote **11** are inducted into the IPM assembly **10**, the tote delivery conveyor **20** automatically advances the tote **11** to the tote recovery conveyor **22**, and a new tote **11** is delivered to the IPM assembly **10** via the tote delivery conveyor **20**. In an embodiment shown in FIG. 2, the chute **24** includes guide walls **26** that guide each tote **11** toward the tote recovery conveyor **22** positioned in a plane below the tote delivery conveyor **20**. In an embodiment, the chute **24** is curved towards the tote recovery conveyor **22**, and each tote **11** slides down the chute **24** to the tote recovery conveyor **22** that returns the totes **11** for reuse in the order fulfillment system.

[0051] Referring to FIGS. **1a** and **1b**, the IPM assembly **10** includes components or assemblies that carry out a sequence of operations to fulfill an order, including an item introduction or induction assembly **30** that indexes a position of each item or product **12** introduced into the IPM assembly **10**, a verification assembly or label application assembly **40** that images and verifies each item **12** and may include an operation to apply patient-specific information to the item **12**, such as via a label **45**, and a packaging assembly **50** that performs one or more of the following operations: verifying and packaging of each item **12**, printing and packaging patient information documents, if required, that corresponds to each item **12** and/or each order, and printing order identification information on the packaging, including order identification indicia that corresponds to the order and/or address information, such as a patient name and an order ship-to address. Each component is modular, meaning that each component is a self-contained unit, and may be arranged flexibly relative to the order fulfillment system to carry out the corresponding operation.

[0052] The IPM assembly **10** includes an item indexing conveyor **52** that conveys or advances each item **12** between the components of the IPM assembly **10**. The operations of the IPM assembly **10** are coordinated, and the item indexing conveyor **52** advances when the operation in each of the item induction assembly **30**, label applicator assembly **40**, and packaging assembly **50** are completed in each cycle. In an embodiment, a conveyor housing **55** extends in covering relationship with the item indexing conveyor **52**. As best shown in FIG. 3, the item indexing conveyor **52** includes a base **100** and an edge mounted conveyor **110**. In an embodiment, the base **100** forms a conveyor run **114**, which extends from at least the item induction assembly **30** to at least the label applicator assembly **40** and the packaging assembly **50**, and items **12** are driven or advanced on the conveyor run **114** by a plurality of consecutively aligned vee- or V-shaped blocks **148**. The edge mounted conveyor **110** projects vertically upward from the base **100** and relative to the base **100** and rotates around a drive roller **111** at one end and an idler roller **112** at an opposite end. The plurality of V-shaped blocks **148** are coupled to the edge mounted conveyor **110**, spaced above the base **100** such that as the edge mounted conveyor **110** is driven, the plurality of V-shaped blocks **148** advance over the base **100** and completely around the edge mounted conveyor **110**. The base **100** may include one or more openings **149** (see FIGS. 8 and 12) through or below which an assembly for an operation may be installed.

[0053] Each of the plurality of V-shaped blocks **148** is able to engage an item **12** introduced or inducted into the IPM assembly **10** and advance the item **12** on an operations side or front or first longitudinal side **150** of the conveyor run **114** from the item induction assembly **30** proximate a first end **154** to the packaging assembly **50** proximate a second end **158**. In an embodiment shown in FIG. 3, after each item is advanced into the packaging assembly **50**, each of the plurality of V-shaped blocks **148**, empty and without an item **12** retained thereby, is conveyed around the second end **158**, along a rear or second longitudinal side **164**, and then around the first end **154** of the item indexing conveyor **52**. Driving means **172**, such as a hydraulic motor **172** or another conveyor driver known in the art, is operably coupled to the drive roller **111** for driving the edge mounted conveyor **110** to which the plurality of V-shaped blocks **148** are coupled.

[0054] FIG. 4 shows an embodiment of a V-shaped block **148** that is configured such that a

plurality of V-shaped blocks **148** are consecutively positionable to advance indexed items **12** along the conveyor run **114** that are inducted into or placed in the IPM assembly **10**. Each of the plurality of V-shaped blocks **148** is formed with a retaining notch **178** in which one item **12** may be positioned, a leading side **182**, a trailing side **188**, and an inner side **190** that is coupled to the edge mounted conveyor **110**. The trailing side **188** of each leading V-shaped block **148** is positioned proximate to the leading side **182** of each following V-shaped block **148**. The retaining notch **178** of each of the plurality of V-shaped blocks **148** is formed between or defined by a drop surface **192** and a pushing surface **194**. The drop surface **192** is angled relative to the pushing surface **194** to retain an item **12** in the retaining notch **178**, such that as each of the plurality of V-shaped blocks **148** with the item **12** retained in the retaining notch **178** is driven along the first longitudinal side **150**, the item **12** does not rotate relative to the direction of conveyance of the item indexing conveyor **52** or move laterally relative to the V-shaped block **148** in which the item **12** is retained. In the embodiments shown in FIGS. **1a**, **1b**, and **3**, the second longitudinal side **164** of the item indexing conveyor **52** is lower than the first longitudinal side **150**, or the base **100** is angled downward relative to the first longitudinal side **150**, such that an item **12** placed in the retaining notch **178** moves, through gravity, to a position in the retaining notch **178** that is adjacent to or proximate an intersection between the drop surface **192** and the pushing surface **194**.

[0055] The item induction assembly **30** facilitates induction of each item into the IPM assembly **10**, which includes retrieving order information from a database regarding each order, indexing a position of an item on the item indexing conveyor **52**, directing placement of each item in a predetermined orientation on the item indexing conveyor **52**, communicating labeling information and item identification information of each item **12** to the label applicator assembly **40**, and communicating packaging information of each item **12** to the packaging assembly **50** that includes a document printing station. Indexing includes recording a position of the item **12** on the item indexing conveyor **52**, or recording a position of the item **12** relative to other items **12**, and communicating the indexed position to the label applicator assembly **40** and the packaging assembly **50** for verification of the item **12** as the item **12** advances through the label applicator assembly **40** and the packaging assembly **50**.

[0056] In an embodiment shown in FIG. **6a**, the item induction assembly **30** includes an indexing cycle indicator **200**, an optical imaging device **204**, the processing system **15** with a monitor **214**, and a sensor **220** for sensing a presence of each item introduced onto the item indexing conveyor **52**. In an embodiment shown in FIG. **6b**, the item induction assembly **30** is similar to the item induction assembly **30** shown in FIG. **6a**, and also includes a housing **223** that supports the monitor **214** in close proximity to the item indexing conveyor **52** and a light curtain **218** that triggers operation of the item indexing conveyor **52**.

[0057] In an embodiment shown in FIG. **6a**, the indexing cycle indicator **200** provides at least two signals for indicating a stage of the item indexing conveyor **52**. The indexing cycle indicator **200** may include a light indicator that changes color to indicate, to an operator, in which stage or mode the indexing conveyor **52** is operating. For example, a red light may be used to indicate that the item indexing conveyor **52** is in an indexing cycling mode or advancing and that items are not introducible into the IPM assembly **10**. A yellow light may be used to indicate that the indexing cycling mode is ending within a predetermined period. A green light may be used to indicate that the item indexing conveyor **52** is in a loading mode or paused and that items may be introduced into the IPM assembly **10**. In an embodiment, the light indicator of the indexing cycle indicator **200** may be a continuous, pulsed, and/or staccato light. The indexing cycle indicator **200** is communicatively coupled to the processing system **15** and supported on the processing system **15** adjacent the monitor **214**, and an operator is able to observe the indexing cycle indicator **200** and the monitor **214** approximately simultaneously and while imaging and introducing each item **12** into the IPM assembly **10**. As used herein, it is foreseeable that an operator may be a robotic tool, or another technology known in the art, that is able to perform tasks without human intervention.

[0058] In an embodiment shown in FIG. 6b, the light curtain 218 is used instead of the indexing cycle indicator 200, and the light curtain 218 triggers operation of the item indexing conveyor 52, including advancement of the item indexing conveyor 52 and pausing the advancement of the item indexing conveyor 52. The light curtain 218 includes a transmitter that emits a plurality of light beams and receiver that detects the light beams. Interruption of the light beams by an operator triggers the item indexing conveyor 52 to pause, which enables the operator to place the item 12 onto the item indexing conveyor 52. Uninterruptedness or continuity of the light enables the item indexing conveyor 52 to advance the item indexing conveyor 52 and the items 12 contained therein. It is foreseeable that an autonomous robot may be used to place each item 12 onto the item indexing conveyor 52.

[0059] In an embodiment shown in FIG. 2, induction commences when the identification tag reader 14 coupled to a bottom of a frame of the item indexing conveyor 52 and directed toward and within read range of the identification tag 13 on the tote 11 when the tote 11 is conveyed into alignment with or right before an opening 221 in the conveyor housing 55 activates the identification tag reader 14. After the identification tag 13 is read by the identification tag reader 14, an item identification indicia or a product identification label (i.e. a barcode) 222 on each item 12 in the tote 11 is imaged by the optical imaging device 204 just prior to the item 12 being placed on the item indexing conveyor 52. In an embodiment, the optical imaging device 204 images or captures the barcode 222 of item 12 as the operator removes the item 12 from the tote 11. The optical imaging device 204 is communicatively coupled to the processing system 15, which identifies the item 12 by comparing the barcode 222 of the item 12 to the item identification information of the order information. The order information includes an orientation that the item 12 is to be positioned relative to the V-shaped block 148, and a signal or prompt is provided by the processing system 15 for the operator detailing the orientation of the item 12. In an embodiment, the prompt is an operator visual aid (“OVA”) shown via the monitor 214, and the OVA includes a barcode loading orientation information view, such as one of the views shown in FIGS. 5a-5g, that details a direction that the barcode 222 on the item 12 is to be directed relative to the retaining notch 178 of the V-shaped block 148 when the item 12 is placed therein. The OVA may also include other product information or a product image for verification of the item 12 by the operator.

[0060] Each item is placed on the item indexing conveyor 52 through the opening 221 in the item indexing conveyor housing 55. In an embodiment shown in FIG. 6a, the monitor 214, which may be a touchscreen monitor, is positioned above the opening 221 to facilitate viewing the barcode loading orientation information and placing the item 12 on the item indexing conveyor 52. In an embodiment shown in FIG. 6b, the item induction housing 223 supports the monitor 214 adjacent to the item indexing conveyor 52 that extends through the item induction housing 223 to facilitate imaging the item 12 with the optical imaging device 204, viewing the barcode loading orientation information, and placing the item in the retaining notch 178 of the V-shaped block 148 on the item indexing conveyor 52. The operator is able to view the monitor 214 and the V-shaped block 148 approximately simultaneously or with minimal movement or rotation of the operator's head. In an embodiment, the item induction housing 223 is positioned over the item indexing conveyor 52 and at a first end 154 thereof and supports the monitor 214 adjacent the rear side 164 conveyor housing 55 such that an operator views the monitor 214 by looking over and across the item indexing conveyor 52.

[0061] As shown in FIGS. 5a-5f, a six-sided polygonal item 12 may be loaded into the retaining notch 178 of one of the plurality of V-shaped blocks 148 in one of six different orientations: FIG. 5a shows the item 12 loaded with the barcode 222 directed toward the drop surface 192; FIG. 5b show the item 12 loaded with the barcode 222 directed upward; FIG. 5c shows the item 12 loaded with the barcode 222 directed away from the pushing surface 194; FIG. 5d shows the item 12 loaded with the barcode 222 directed away from the drop surface 192; FIG. 5e shows the item 12 loaded with the barcode 222 directed downward; and FIG. 5f shows the item 12 loaded with the

barcode **222** directed toward the pushing surface **194**. FIG. **5g** shows an item **12** that is a cylindrical container with the barcode **222** directed away from the pushing surface **194**, similar to the polygonal item **12** shown in FIG. **5c**. In an embodiment items of other shapes may be loaded into a V-shaped block **148**, and the OVA will display the barcode loading orientation information of the item **12** relative to the V-shaped block **148** as each item is inducted into the item induction assembly **30**.

[0062] The sensor **220** is a beam sensor that is positioned to detect at least a presence of the item **12** placed onto the item indexing conveyor **52**, verifying that the item **12** is being processed in the IPM assembly **10**. In an embodiment shown in FIG. **6a**, the sensor **220** is secured to a sensor support **224** that is coupled to a frame of the item indexing conveyor **52** and extends therefrom such that the sensor **220** is positioned above the item **12** that is placed in the retaining notch **178** of a V-shaped block **148** positioned to receive the item **12**. As best shown in FIG. **6a**, a sensing path or beam **230** of the sensor **220** is directed along or adjacent to the drop surface **192** of the V-shaped block **148** to detect the item **12** and verify placement of the item **12** in the item indexing conveyor **52**. The sensor **220** is communicatively coupled to the processing system **15**, and a position of the item **12** is indexed or associated with a V-shaped block **148** upon placement of the item **12** in the item indexing conveyor **52**. Upon induction and verification of the item **12** into the item induction assembly **30**, the processing system or director **15** communicates labeling information to the label applicator assembly **40**, packaging information to the packaging assembly **50**, and queue or indexing information of the item **12** to the label applicator assembly **40** and the packaging assembly **50**.

[0063] In an embodiment shown in FIG. **6a**, the opening **221** of the conveyor housing **55** may include an opto-electronic or presence detection device **240** that is able to detect a presence of an object or user and is connected to a safety relay device that is enabled to remove motive power from the item indexing conveyor **52** when an object is detected. The opto-electronic device **240** includes a transmitter and a receiver, one of which is positioned above the opening **221** and another of which is positioned below the opening **221**. In an embodiment, the opto-electronic or presence detection device **240** is selectively operable to function during the indexing cycling mode. When an object is sensed between the transmitter and receiver of the presence detection device **240**, advancement of the V-shaped blocks **148** of the item indexing conveyor **52** is stopped or paused and prevented from moving until the object is removed. Presence detection devices **240** are commonly used safety devices known by those skilled in the art.

[0064] In an embodiment shown in FIG. **6b**, a stop device **250** is coupled to the item indexing conveyor **52** to stop or pause advancement of the item indexing conveyor **52**, which may be utilized if the light curtain **218** fails to trigger the item indexing conveyor **52** to stop advancement. The stop device **250** may be a switching device or another safety device known in the art.

[0065] Referring to FIGS. **7-10b**, the label applicator assembly **40** includes: a lowerator platform or item receiver platform **300** that is moveably supported by a lowerator housing or item receiver housing **302** and moveable relative to the base **100** of the item indexing conveyor **52** or a first or upper position (shown in FIG. **7**) to a second or lowered position (FIG. **8**); a set of cameras **305a-305f** (FIG. **9**) that forms an imaging station; and a label applicator **310** moveably supported in a superstructure **312** (FIG. **10a**) or in a mini-structure **313** (FIG. **10b**). The labeling information received via the processing system **15** may include a patient-specific information print file that includes customer identifying indicia and patient-specific information that is to be applied to the item **12**, which may also be referred to as customer information, label application location information indicating a location of where the patient-specific information is to be applied relative to a surface of the item, and label position information indicating a labeling position of the lowerator platform **300** relative to the base **100** where the patient-specific information is to be applied to the item **12**. In an embodiment, the labeling position is a position of the lowerator platform **300** where verification and/or application of the label **45** occurs and may be in the first

position or the second position. In an embodiment, the label applicator assembly **40** receives the labeling information for each item **12** and temporarily stores the labeling information for the item **12** until a previous item is advanced from the label applicator assembly **40**, and the label **45** is printed simultaneously or just before the item **12** is positioned on the lowerator platform **300**.

[0066] Each of the plurality of V-shaped blocks **148** of the item indexing conveyor **52** sequentially advances adjacent to the lowerator platform **300**. As shown in an embodiment in FIGS. 7-9, the base **100** of the item indexing conveyor **52** includes an opening **149** extending therethrough sized to correspond to at least the lowerator platform **300**. As the plurality of V-shaped shaped blocks **148** are advanced around the edge mounted conveyor **110**, each retaining notch **178** of the plurality of V-shaped blocks **148** is conveyed into alignment with the lowerator platform **300** positioned in the upper position in the lowerator housing **302** and in the opening **149** of the base **100**. In the upper position, the lowerator platform **300** is approximately aligned with the base **100** of the item indexing conveyor **52**, and when the V-shaped block **148** is conveyed into alignment with the lowerator platform **300**, an item **12** retained in the retaining notch **178** of the V-shaped block **148** is advanced onto the lowerator platform **300**.

[0067] Upon arrival on the lowerator platform **300** of the label applicator assembly **40**, the item **12** is verified. Verification of the item **12** may require imaging of one or more of the surfaces of the item **12**, and preferably includes imaging at least the surface with the barcode **222** of the item **12**. In an embodiment, at least one camera of the set of cameras **305a-305f** is used to image the barcode **222** of the item **12**. The barcode **222** of the item **12** is compared to at least one of: the item identification information of the order information to verify that the item **12** is the same as an ordered item of the order, an expected item in the indexed position, and/or the item **12** identified on the printed label **45**. If the item **12** is verified, the label **45** is applied to the item **12**. If the item **12** is not verified, the item **12** is rejected and the label **45** is not applied to the item **12**, as described herein.

[0068] In an embodiment, upon arrival of the item **12** on the lowerator platform **300** at least one camera of the set of cameras **305a-305f** captures images of one or more surfaces of the item **12** prior to application of the label **45** on the item **12**. The at least one image may be used in a pharmaceutical verification and compared to at least a third party or pre-existing image that corresponds to the item identification information of the order information.

[0069] In an embodiment, a platform elevator **315** functions to move the lowerator platform **300** within the lowerator housing **302**. The lowerator platform **300** is mounted on a platform frame structure **318** that extends from a bottom surface of the lowerator platform **300**. The platform elevator **315** includes a carriage **320** coupled to the platform frame structure **318**, a rail **322** that provides a track or channel **324** in which the carriage **320** is movably secured, a drive assembly that is actuatable to move the lowerator platform **300** from the upper position to the lowered position and then return the lowerator platform **300** to the upper position, and a platform elevator support member **325**.

[0070] In an embodiment, each item **12** received on the lowerator platform **300** is verified and/or imaged prior to labeling of the item **12**, as described herein. The initial verification and imaging at the label applicator assembly **40** may occur when the lowerator platform **300** is in the upper position or in the lowered position. Preferably, verification and imaging occurs in the same position as the application of the label **45** to the item **12** and/or post-labeling imaging. When the lowerator platform **300** is in the upper position, cameras **305a-305d** are in an upper position and are able to read or image one or more of the surfaces of the item **12** directed toward the cameras **305a-305d**, including the surface with the barcode **222** as shown in FIGS. 5b-5e. Alternatively, when the lowerator platform **300** is in the lowered position, cameras **305a-305d** are in a lowered position and cameras **305a-305f** are able to read or image one or more of the surfaces of the item **12** directed toward the cameras **305a-305f** including the surface with the barcode **222** as shown in FIGS. 5a-5f. In the lowered position, cameras **305e** and **305f** are able to read or image the respective surfaces of

the item **12** shown with a barcode **222** in FIGS. **5a** and **5f**, which are not able to be imaged when the lowerator platform **300** is in the upper position.

[0071] Cameras **305a-305d** are secured to a camera elevator **328** that positions the cameras **305a-305d** in the upper position or in the lowered position corresponding to the position of the lowerator platform **300**. Cameras **305e** and **305f** are fixed to the lowerator housing **302** and are stationary. Each of the cameras **305a-305f** is directed toward one surface or side of the item **12** on the lowerator platform **300**. When the lowerator platform **300** is in an upper position, the cameras **305a-305d** are in an upper position: camera **305a** is directed toward and images an upper surface of the item **12**; camera **305b** is directed toward and images a surface of the item **12** that is directed away from the drop surface **192**; camera **305c** is directed toward and images a surface of the item **12** that is directed away from the pushing surface **194**; and camera **305d** is directed toward a bottom surface of the item **12** and images the bottom surface through the lowerator platform **300**. Cameras **305a-305d** are lowered to a lowered position when the lowerator platform **300** is lowered and image the respective surface of the item **12** as described relative to the upper position. Cameras **305e** and **305f** image respective surfaces of the item **12** when the lowerator platform **300** is in the lowered position. Specifically, camera **305e** is directed toward a surface of the item **12** that was directed toward the drop surface **192** in the upper position and images the surface through a first housing wall **330** (see FIG. **8**) of the lowerator housing **302**; and camera **305f** is directed toward a surface of the item **12** that was directed toward the pushing surface **194** in the upper position and images the surface through a second housing wall **335** (see FIG. **8**) of the lowerator housing **302**. [0072] At least a portion of the lowerator platform **300**, the first housing wall **330**, and the second housing wall **335** are configured such that the respective cameras **305d**, **305e**, and **305f** are able to image the respective surface of the item. Preferably, the lowerator platform **300**, the first housing wall **330**, and the second housing wall **335** are transparent and formed from a material, such as, for example, a glass, an acrylic, a polycarbonate, another transparent material known in the art, or a combination of such materials.

[0073] The lowerator housing **302** comprises a base **340** having a lower surface to which a housing frame **345** is secured. An upper surface of the base **340** of the lowerator housing **302** is configured to be secured to a bottom surface of the indexing conveyor **52**. As shown in FIG. **9**, a plurality of securing holes **348** extend into the upper surface of the base **340** for securing the lowerator housing **302** to the item indexing conveyor **52** such that the lowerator platform **300** is in planar alignment with the base **100**.

[0074] The housing frame **345** of the lowerator housing **302** extends below the base **340** and supports mounts **350** to which the cameras **305e** and **305f** are mounted or secured. The housing walls **330** and **335** also extend below the base **340** and are supported by the housing frame **345**. The lowerator platform **300** moves relative to the housing walls **330** and **335**, and the housing walls **330** and **335** extend between the base **340** and the lowerator platform **300** when the lowerator platform **300** is in the lowered position such that the surface of the item **12** that was directed toward the drop surface **192** in the upper position and the surface of the item **12** that was directed toward the pushing surface **194** in the upper position are viewable through the respective housing wall **330** and **335** and imageable by the respective camera **305e** and **305f**. The lowerator housing **302** may be supported by one or more support legs **352** secured to the superstructure **312** and/or the mini-structure **313**.

[0075] The camera elevator **328** that supports cameras **305a-305d** includes a rail **370** that provides at least one track or channel **382** for adjustably mounting each of the cameras **305a-305d** thereto. The camera elevator **328** is vertically moveable with the lowerator platform **300** and functions to move the cameras **305a-305d** with the lowerator platform **300** as the lowerator platform **300** is moved to the upper position and the lowered position. In an embodiment, the camera elevator **328** is secured to the housing frame **345** of the lowerator platform **300** or to another portion of the lowerator platform **300**. It is foreseeable that the camera elevator **328** may be coupled to the

platform elevator **315**. It is contemplated that the camera elevator **328** may be vertically moveable independent of the lowerator platform **300**.

[0076] In the embodiment shown in FIG. **9**, the rail **370** has at least one channel **382** formed on at least three of the sides thereof, and each of the cameras **305a-305d** is securely mounted to the rail **370** in one of the channels **382** by a mounting system **384** that includes a mounting base **386** that is adjustably securable in the channel **382** and one or more two-way connectors **390** adjustably securable to one or more rods **392** that extend between the mounting base **386** and a camera bracket **394** secured to one of the cameras **305a-305d**. The mounting base **386** is vertically adjustable relative to the channel **382** to adjust a vertical position of the cameras **305a-305d**. The one or more two-way connectors **390** provide adjustability of a distance of each camera **305a-305d** from the mounting base **386** and rotational adjustability to further adjust a position or an orientation of each camera **305a-305d**. In an embodiment, each of the cameras **305a-305d** are independently moveable relative to the camera elevator **328**.

[0077] After the item **12** is verified, a label **45** may be applied to the item **12** while the lowerator platform **300** is in one of the upper or the lowered position as indicated by the processing system **15**. The label applicator **310** is supported by the superstructure **312** (see FIG. **10a**) or mini-structure **313** (see FIG. **10b**) and communicatively connected to the processing system **15**. As described herein, the labeling information sent from the processing system **15** may include a patient-specific information print file such that a label **45** is printed in the label applicator assembly **40** for the item **12**, label application location information instructions for positioning the label **45** relative to a surface of the item **12**, and label position information that includes whether application of the label and subsequent verification occur with the lowerator platform **300** in the upper position or in the lowered position. The label applicator **310** is configured such that the patient-specific information from the processing system **15** is printed on a label **45** and ready for application when the corresponding indexed item **12** is on the lowerator platform **300** in the labeling position.

[0078] As shown in FIGS. **10a** and **10b**, the label applicator **310** includes a printer assembly **402** that prints a label **45** with patient-specific information corresponding to an indexed item **12** and a tamp or label applicator head **404** that applies the printed label **45** to the item **12**. A motion control assembly **406** adjustably positions the tamp **404** relative to the item, adjustably moving the label applicator **310** to position the tamp **404** such that the printed label **45** is applied to a surface of the item **12** in a location on the item as communicated by the processing system **15**, preferably avoiding identification labels or barcodes, a national drug code, a stock-keeping unit barcode, an expiration date, or other identifying information known in the art. A contact surface **410** of the tamp **404** extends approximately parallel to the drop surface **192** of the V-shaped block **148** when the lowerator platform **300** is in the upper position or extends approximately parallel to the first housing wall **330** when the lowerator platform **300** is in the lowered position.

[0079] The printer assembly **402** is known in the art and includes printing technology that is applicable to printing labels **45**. In an embodiment, a direct thermal label printer applies heat to a label **45** having a heat-sensitive coating on a first side to print the patient-specific information thereon. A second side of the label **45** has an adhesive coating on at least a portion thereof to maintain the label **45** in a folded configuration and to adhere the label **45** to a corresponding item. The printed labels **45** are advanced from the printer assembly **402** to a transferring or folding mechanism **420** such that the tamp **404** is able to receive the printed label **45** with the adhesive coating facing away from the contact surface **410** of the tamp **404**. The folding mechanism **420** is in a label receiving or open position. The folding mechanism **420** is actuated into a closed position to fold the printed label **45** onto the tamp **404**, and the folded printed label **45** is adhered by a portion of the adhesive coating and leaves an exposed adhesive area that is applicable to a selected label location on an item **12**. In an embodiment, the exposed adhesive area is approximately 3/8-inches along one side of the folded label **45**. The folding mechanism **420** is returned to the open position and the tamp **404** is advanced forward with the folded printed label **45** thereon such that at

least the exposed adhesive area of the folded label **45** on the contact surface **410** of the tamp **404** engages a surface of the item **12** or the label location that corresponds to the label application location of the label **45** indicated by the processing system **15**.

[0080] In the superstructure **312** and the mini-structure **313**, the positioning of the tamp **404** is movable by the motion control assembly **406** which may include one or more positioning tracks. The positioning tracks may include one or more of: one or more first tracks **425** that moves the tamp **404** closer to or further from the item indexing conveyor **52**, a second track **430** that moves the tamp **404** horizontally in a plane that is approximately parallel to the plane of the drop surface **192**, and a third track **440** that moves the tamp **404** vertically in a plane that is approximately parallel to the plane of the drop surface **192**. The second track **430** is secured to a first carriage assembly **450** that is slidably securable on the one or more first tracks **425**, and the third track **440** is secured to a second carriage assembly **455** that is slidably securable on the second track **430**. The label applicator **310** is secured to a third carriage assembly **460** that is slidably securable on the third track **440**. An embodiment of the superstructure **312** is described herein, but variants of the superstructure **312** would be understood by one skilled in the art to function similarly. For example, the mini-structure **313** is a variant of the superstructure **312** that has been flipped or inverted to minimize a footprint of the label applicator assembly **40** and to facilitate maintenance of the label applicator assembly **40** and functions similarly to the superstructure **312**.

[0081] In the embodiment of the superstructure **312** shown in FIG. **10a**, each of the tracks **425**, **430**, and **440** includes one or more rails **465** that guide the corresponding carriage assemblies **450**, **455**, and **460**. In the embodiment shown, each track **425**, **430**, and **440** includes two rails **465** spaced a distance apart that are secured to a beam member **468**. The carriage assemblies **450**, **455**, and **460** are formed with one or more channels **469** that correspond to and receive each of the one or more rails **465** and enable the carriage assemblies **450**, **455**, and **460** to slide to position the tamp **404**. The rails **465** extend a length that corresponds to a potential positioning of the label **45** on the item **12** by the tamp **404**. The tracks **425**, **430**, and **440** each include at least two limiting stops **470** that limit the movement of the corresponding carriage assembly **450**, **455**, and **460** on the rails **465**.

[0082] The one or more one or more first tracks **425** are secured across the superstructure **312** and support the label applicator **310**, the printer assembly **402**, and the motion control assembly **406**. The carriage assembly **450** includes one or more channel housings **475** through which the one or more channels **469** in engagement with the one or more rails **465** extend. A support structure **480** is connected to and suspended from the one or more channel housings **475**. The support structure **480** includes one or more spacers **482** that are secured to the one or more channel housings **475** and a frame **485** that supports a first support bracket **492** and a second support bracket **495** between which the second track **430** is positioned and secured. The support brackets **492** and **495** are sized such that the second track **430** is positionable at an angle to position the contact surface **410** of the tamp **404** parallel to the drop surface **192** of the V-shaped block **148** aligned with the lowerator platform **300**.

[0083] In an embodiment, the tamp **404** has a first home position that is spaced a distance of approximately ¼-inch to the left of the pushing surface **194** of the V-shaped block **148** and ¼-inch above the lowerator platform **300** in the upper position, and a second home position that is spaced a distance of approximately ¼-inch to the left of a plane extending through and below the pushing surface **194** of the V-shaped block **148** and ¼-inch above the lowerator platform **300** in the lowered position. The tamp **404** may be adjustably moved relative to the first and second home positions in a horizontal direction and/or in a vertical direction for application of the label **45** with the patient-specific information to the item **12** in the selected location and position. Directional references, such as left, right, above, and below, used in the description of movement of the tamp **404** or the motion control assembly **406** herein are with reference to a direction of the tamp **404** from a viewpoint of a user facing the operations side **150** of the IPM assembly **10** and movement of the tamp **404** in a plane parallel to the drop surface **192** of the V-shaped block **148**. Further, as used

herein and with regard to the IPM assembly **10**, references to horizontal may refer to a plane or axis parallel to the lowerator platform **300** or the base **100** or the conveyor run **114**, and references to vertical may refer to a plane or axis parallel to the drop surface **192** of the V-shaped block **148** or a plane perpendicular to at least the conveyor run **114**.

[0084] As described, the third track **440** enables the tamp **404** to apply the label **45** to the item **12** when the item **12** is on the lowerator platform **300** in the upper position or in the lowered position. The third track **440** also enables the tamp **404** to apply the label **45** in a selected vertical position on a surface of the item that is directed away from the drop surface **192** and directed toward the contact surface **410** of the tamp **404**. In an embodiment, a vertical position of the tamp **404** has a vertical range in a plane parallel to the drop surface **192** of the V-shaped block **148** of up to 12 inches. It is foreseeable that the tamp **404** could have a vertical range in a plane parallel to the drop surface **192** of the V-shaped block **148** of up to 24 inches. In an embodiment, the tamp **404** is moveable in the vertical direction up to four inches above or up to eight inches below a horizontal plane extending parallel to the lowerator platform **300** in the upper position and through the first home position.

[0085] The second track **430** enables the tamp **404** to apply the label **45** in a selected horizontal position on a surface of the item **12** that is directed away from the drop surface **192** and directed toward the contact surface **410** of the tamp **404**. In an embodiment, the horizontal position of the tamp **404** has a lateral or horizontal range in a plane parallel to the lowerator platform **300** of up to 12 inches. In an embodiment, the tamp **404** is moveable in the horizontal direction up to 12 inches to the left of a vertical plane extending through the home position.

[0086] As shown in FIGS. **11a-11d**, the label **45** applied by the tamp **404** may be applied to items **12** of a variety of different shapes and sizes by varying a horizontal position and/or a vertical position of the tamp **404**. For example, to apply a label **45** to a flat or a rectangular prism or box-shaped item **12** with six surfaces, the tamp **404** may be positioned in the home position or another position to the left and/or above the home position that corresponds to a surface of the item. In FIGS. **11b** and **11c**, rounds or cylindrically-shaped items **12** are shown and the tamp **404** is positioned relative to a dimension of the item for application of the folded printed label **45** onto the item **12**. In an embodiment, the tamp **404** is positioned a distance from the pushing surface **194** of the V-shaped block **148** such that the exposed adhesive area of the label **45** on the tamp **404** is applied to the cylindrically-shaped item **12** at approximately a dimension of $\frac{1}{2}$ a distance of the diameter of the item from the pushing surface **194**. The location of the label **45** on each item **12** may be varied horizontally and vertically by changing a position the tamp **404**, horizontally and vertically, such as, for example, to avoid applying the label **45** onto identifying labels or barcodes of the item, and may include overhanging the label **45** on an edge of the item **12**.

[0087] As shown in FIG. **7**, a tamp pad **498** is positioned on the platform elevator support member **325** for instances in which the label **45** is not applicable to the item **12**. For example, such instances may include when the item **12** on the lowerator platform **300** or upon arrival at the label applicator assembly **40** is not verified as described herein and does not match the label **45**. In such a case, the item **12** becomes an exception and the label **45** is not applied to the item **12**. The label applicator **310** may be manually or automatically paused or put in a stop state or mode, and the label **45** may be manually removed from the label applicator **310** by an operator or the tamp **404** may be lowered or raised to and aligned with the tamp pad **498** such that the label **45** is applicable to the tamp pad **498**. The item **12** may be thereafter advanced through the packaging assembly **50** to a station for manual labeling.

[0088] After the folded, printed label **45** is applied to each item **12**, the item **12** is imaged by at least camera **305b** that is directed toward and captures an image of the surface of the item **12** with the applied label **45**. In an embodiment, an image of the surface of the item **12** with the barcode **222** is simultaneously imaged. The images captured after the item **12** is labeled may be referred to as verification images, and in an embodiment the verification images are captured with the lowerator

platform **300** in the same position as where the label **45** was applied to the item **12**. One or more of the cameras **305a-f** may capture one or more images of one or more of the surfaces of the item **12**. [0089] In an embodiment, a pharmaceutical verification process is performed between capturing the verification images with the label **45** applied to the item **12** and the item **12** arriving at an item verification station of the packaging assembly **50**. A pharmaceutical verification process comprises creating a file that includes a predetermined set of images **499** of the item **12** that are captured in the label applicator assembly **40** and compared by an operator or pharmacist to verify the item **12** that was labeled therein corresponds to the item identification information of the ordered item in the order information. The images used for verification may include one or more images of one or more of the surfaces or sides of the item **12**, including the image with the label **45** applied to the item **12** and the image of the barcode **222** on the item **12**. Verification may occur by an operator or pharmacist that is on-site or at a remote site. In an embodiment, the operator receives the set of images **499** captured in the label applicator assembly **40** and the order information on a monitor or personal computing device, compares the set of images **499** and the order information, and confirms or excepts the labeled item **12** using an input device. In an embodiment shown in FIG. **15**, the set of images may be arranged together for comparison and includes a captured image of a front or surface of the item **12** in the label applicator assembly **40** and prior to labeling, a third party or pre-existing image that corresponds to the item identification information of the ordered item in the order information, the captured image of the surface of the item **12** with the label **45** applied thereto, and the captured image of the surface of the item **12** with the barcode **222** that is captured simultaneously with the image with the label **45** applied to the item **12**. In an embodiment, the label **45** and/or the barcode **222** on the item **12** are highlighted or emphasized and compared during the pharmaceutical verification process to the item identification information. In an embodiment, the set of images **499** are compared by a processing system or automated program, which may utilize software, machine learning, and/or artificial intelligence. In an embodiment, the pharmaceutical verification process may occur after each of the orders is packaged.

[0090] The item **12** is advanced from the labeling position of the lowerator platform **300** and toward the packaging assembly **50**. If in the lowered position, the lowerator platform **300** with the item **12** supported thereon is moved to the upper position. From the upper position, the item **12** is advanced in its indexed position by the corresponding V-shaped block **148** on the base **100** toward the packaging assembly **50**.

[0091] Referring to FIG. **12**, the packaging assembly **50** includes a verification station **500** that images the label **45** on each item **12** for verification of the item **12**, a chute or funnel **505** through which at least one item **12** of an order is advanced into a package **615** having order identification information **510** (see FIG. **13b**) applied thereon that corresponds to the order, a document printing station having one or more printers **640** (shown in FIG. **1a**) that are able to print and deliver patient information documents associated with the item or items **12** in each order for inclusion in the package, and a conveyor **650** that is able to advance a sealed package **615** with the order therein to exit the IPM assembly **10** or reject the package **615** and advance the package **615** to a review conveyor **665** to be reviewed by an operator.

[0092] The packaging assembly **50** receives the packaging information from the processing system **15** for each order and for each item **12** in the order. The packaging information may include indexing information for each item **12** in the order, a patient information document print file, if required, for printing the patient information documents in the document printing station, and order identification information for applying to each package.

[0093] In an embodiment, the verification station **500** comprises a presence sensor and/or a packaging assembly imaging device **530** secured to a support frame **535** of the item indexing conveyor **52**. The packaging assembly imaging device **530** is in alignment with an opening **540** in the conveyor housing **55** to detect when the item **12** is advanced into the verification station **500** and/or to image the item **12** retained in the V-shaped block **148** that is advanced into a position that

corresponds to the opening **540**. The packaging assembly imaging device **530** is activated to image the label **45** of the item **12** through the opening **540** to compare the label **45** to the item identification information in the order information to verify the item **12**.

[0094] As shown in FIG. **12**, subsequent to the verification station **500** the opening **149** in the base **100** corresponds to the chute **505** coupled to the item indexing conveyor **52**. The opening or drop gate zone **149** extends under one or more of the retaining notches **178** of the V-shaped blocks **148**, and the opening **149** forms an entrance into the chute **505**. A gate **555** is in covering relationship with the opening **149** and in alignment with the base **100** such that the item or items **12** advanced onto the gate **555** are advanced through the opening **149** and into the chute **505** when the gate **555** is rotated into an open position.

[0095] One or more presence sensors **550** are positioned to extend in alignment with and over the a position of one or more retaining notches **178** of the V-shaped blocks **148** that cycle or advance over the gate **555**. The presence sensors **550** are able to detect a presence of an item **12** retained in the retaining notches **178** advancing over the gate **555**. In an embodiment shown in FIG. **12**, the opening **149** is sized to correspond to two retaining notches **178**, and two presence sensors **550** are aligned thereabove in a position to detect each item **12**. Upon detection of at least one item **12** by one of the one or more presence sensors **550**, the gate **555** is opened and the one or more items **12** are advanced or dropped into the chute **505**. It is foreseeable that the opening **149** and the gate **555** may be sized to correspond to a desired number of retaining notches **178** greater than or less than that shown in FIG. **12**.

[0096] The sensors **550** are secured to a sensor support **556** that is coupled to a frame of the item indexing conveyor **52** and extends therefrom such that the sensors **550** are positioned above the item or items **12** that are in the retaining notches **178**. As shown in FIG. **12**, one or more sensing paths or beams **558** of the sensors **550** are directed to detect each item **12**. The sensors **550** may be communicatively coupled to the processing system **15**.

[0097] The processing system **15**, in cooperation with the item indexing conveyor **52**, indexes a position of consecutive items in adjacent retaining notches **178** that are part of the same order and/or consecutive items in nonadjacent retaining notches **178** that are part of the same order, but without items of another order therebetween, for packaging together. Packaging of more than one item in one package may require more than one cycle or advancement of one or more items through the chute **505**. In an embodiment, up to four items in one order may be packaged together. In an embodiment, the number of items packaged together is limited by the size of the package, and a plurality of cycles of advancing one or more items **12** through the chute **505** may be required to fulfill an order.

[0098] The frame **535** supports a hinge or pivoting mechanism **560** that pivots or drops the gate **555** down and out of alignment with the base **100**. The gate **555** extends between the pivoting mechanism **560** and the V-shaped blocks **148**, and the pivoting mechanism **560** may be pneumatically controlled for actuation after at least one item **12** is detected by a presence sensor **550**. The gate **555** is configured such that it extends under each retaining notch **178**, and downward pivoting of the gate **555** relative to the pivoting mechanism **560** causes items supported thereon to drop into the chute **505**.

[0099] As shown in FIG. **13a**, the chute **505** extends between the opening **149** in the base **100** and a bagging assembly **600**. The chute **505** is formed in the shape of a funnel, having at least two sides **605** that form a narrowing channel **608**. It is foreseeable that the chute **505** may also be formed in another shape that guides the items into a package **615**. Items dropping or advancing through the opening **149** are guided or directed through the narrowing channel **608** toward an exit **610** of the chute **505**. The exit **610** corresponds to a top of a package **615** that is supported by the bagging assembly **600** and/or a conveyor **650** (see FIG. **13b**). Each item **12** is advanced through the chute **505** and falls through the exit **610** and into the package **615** positioned thereunder. In an embodiment, at least one side **617** of the chute **505** is transparent such that a chute imaging device

620 (see FIG. 12), supported on the frame **535** and directed toward the transparent side, is able to detect an item **12** that is jammed or stuck within the chute **505** and fails to fall into a bag or package **615** positioned under the exit **610** of the chute **505**.

[0100] The bagging assembly **600** is able to apply or print order identification

[0101] information **510**, such as an order identification indicia and delivery information that includes a customer or patient name and an order ship-to address on a package or on a label that is applied to the package, image the order identification indicia on the package with a package verification imaging device **660** to verify that the package corresponds to the order, and after the item or items **12** are advanced into the package **615**, place or drop the package **615** onto a conveyor **650** that conveys the package **615** to exit the IPM assembly **10**. At least one finger device **630** of the bagging assembly **600** opens at least one side of an opening of the package **615** such that the opening of the package **615** is positioned under the exit **610** of the chute **505**

[0102] The document printing station includes the one or more printers **640** that may print one or more patient information documents associated with one or more items in each order for inclusion with the order in the package. The patient information documents may include patient package inserts, medication guides, instructions for use, and/or other information which may be required or is useful and known by one skilled in the art. Each page of the patient information documents may include a unique document identification indicia and/or order identification information. The one or more printed patient information documents are transferred by a document transfer assembly from the one or more printers **640** and are positioned for insertion into the package when the package is opened by the at least one finger device **630**, before or after the one or more items **12** have been advanced into the package. The document transfer assembly includes a document verification imager **645** that images the document identification indicia on each of the pages of the patient information documents to verify that each of the documents corresponds to the order prior to the patient information documents being transferred into the package. **615**. The document transfer assembly may include an auto-fold assembly that folds and inserts the printed patient information documents through a slot that guides the printed patient information documents into the opened package. A blade or device folds and pushes the printed patient information documents into the slot such that the printed patient information documents are advanced into the opened package.

[0103] In an embodiment, the conveyor **650** is a bi-directional conveyor, and each package is dropped onto the conveyor **650** such that the order identification indicia on the package faces upward or away from the conveyor **650**. The package verification imaging device **660** that images the order identification indicia on the package is communicatively coupled to the processing system **15** that is able to selectively determine if a conveyor run **670** of the conveyor **650** is advanced in a first direction such that the package exits the IPM assembly **10** or if the conveyor run **670** of the conveyor **650** is advanced in a second direction such that the item or items contained in the package are reviewed. If the package **615** is verified, the package **615** is sealed after the one or more verified items **12** and/or the one or more verified patient information documents are advanced therein, and the verified package **615** is dropped onto the conveyor **650** and advanced on in the first direction. Items or orders that were unverifiable or previously excepted or rejected may be advanced in the second direction for review. Imaging device **680** is a package sensor that is able to determine when the package **615** has advanced from the conveyor run **670** of the conveyor **650**.

[0104] FIGS. 14a, 14b and 14c are flow diagrams of an embodiment of an item packing machine process **800** that is part of an order fulfillment system and includes the steps of inducting, labeling, and packaging items using an embodiment of the IPM assembly **10**. The process **800** may also be referred to as a method of fulfilling a plurality of orders in the IPM assembly **10**. One or more processing systems or directors **15** that are communicatively coupled to the IPM assembly **10** may be used to execute the process **800** that includes indexing a position of each item **12** on the item index conveyor **52** to facilitate labeling and packaging of the item **12**. Other embodiments of the process **800** may be utilized by the IPM assembly **10**.

[0105] FIG. **14a** shows an embodiment of an induction process **802** for inducing and indexing an item into the IPM assembly **10**.

[0106] At step **804**, a customer sends the order and the order information to an order fulfillment operation that includes the IPM assembly **10**. The order received by the IPM assembly **10** may include, for example, an order from an e-commerce website, an order from a retail service provider, and/or an order from another foreseeable orderer. The order will trigger the order fulfillment system to begin order fulfillment by picking-and-placing the one or more items **12** of each order into the tote **11**.

[0107] At step **806**, each tote **11**, which may contain a multi-item order, a single-item order, or multiple single item orders, is delivered to the IPM tote queue to the IPM assembly **10**.

[0108] At step **808**, the totes **11** are singulated and are advanced to the identification tag reader **14** of the IPM assembly **10**.

[0109] At step **810**, the identification tag **13** on each tote **11** is read by the identification tag reader **14**, such as an RFID reader. The identification tag **13** on each tote **11** corresponds to the one or more orders in the tote **11** and the one or more items **12** of each of the orders in the tote **11**. The identification tag reader **14** is communicatively coupled to the processing system **15**, and the processing system **15** retrieves order information from a database regarding the one or more orders in the tote **11**.

[0110] At step **812**, an operator or item inductor introduces or inducts each item **12** from the order in the tote **11** in succession into the IPM assembly **10** by selecting any one item **12** in the tote **11**, imaging the barcode **222** of the item **12** with the optical imaging device **204** for verification of the item **12** of the order, and placing the item **12** onto the V-shaped block **148** to verify the item **12** as directed or specified by the OVA via the monitor **214** via the director **15** that indicates the orientation of the barcode **222** of the item **12** relative to the V-shaped block **148**.

[0111] At step **814**, the item indexing conveyor **52** indexes the position of the item **12** on the item indexing conveyor **52**.

[0112] At step **816**, the director **15** of the IPM assembly **10** queues packaging information to the packaging assembly **50** for each item **12** indexed. Queuing may include sending one or more of the following: indexed position of item **12**, and order information that includes the print file associated with the item **12** and/or the order for the document printing station **50**. The print file may include order identification information and/or patient information documents for the order.

[0113] During the indexing step, induction indicators are reset for each new item **12** introduced into the IPM assembly **10**. At step **818**, the item indexing conveyor cycles or incrementally advances at least one position as the OVA is reset between each item **12** that is imaged by the optical imaging device **204**.

[0114] At step **820**, the operator or item inductor determines whether one or more items **12** remain in the tote **11** to be inducted into the IPM assembly **10** and indexed on the item indexing conveyor **52**. If one or more items **12** remain, the operator repeats the process from step **812**, inducting the item **12** into the IPM assembly **10** and repeating the steps until the tote **11** is empty. If no items remain of the order in the tote **11**, the tote **11** is advanced to a tote recovery conveyor **22**.

[0115] At step **822**, the director **15** determines whether the one or more orders in the tote **11** have been fulfilled. At step **824**, if one or more orders of the tote **11** are fulfilled, the director **15** of the IPM assembly **10** queues an indexed position of the item **12** and labeling information to the label applicator assembly **40** for each item **12** in the one or more orders. At step **826**, if an order is not fulfilled but one or more items **12** of the order have been indexed on the item indexing conveyor **52**, the order is rejected and any indexed items of the order proceed through the IPM assembly **10** to be manually labeled and verified.

[0116] FIG. **14b** shows an embodiment of an imaging and verification process **830**. In the imaging and verification process **830**, a label may be applied to the item **12**, and an image of one or more sides of the item **12** is captured. The image of the one or more sides of the item **12** is used for

verifying the item **12** in the order.

[0117] At step **832**, the item **12** arrives at the lowerator platform **300**, and the lowerator platform **300** is positioned in a first position with the item **12** in the retaining notch **178** of the V-shaped block **148** or in a second position vertically displaced from the first position.

[0118] In an embodiment, at step **834** one or more sides of the item **12** are imaged, including item identification indicia to verify the item **12**.

[0119] In an embodiment, at step **836** the label **45** is printed with patient-specific information in the printer assembly **402** and applied to the item **12** on the lowerator platform **300** that is in a labeling position. The labeling position of the lowerator platform **300** comprises one of at least the first position and the second position.

[0120] At step **838**, the one or more sides of the item **12** are imaged, including at least a side of the item **12** that includes the label **45**. In an embodiment, the lowerator platform **300** is in the labeling position for the application of the label **45** to the item **12** of step **836**, the imaging of the item identification indicia on the item of step **834**, and the capturing of the one or more images of the item **12** of step **838**.

[0121] At step **840**, the images of the item **12** are used to verify that the item **12** corresponds to an item in the order. In an embodiment, a third party image of the item **12** is compared to the images of the item **12** for verification. In an embodiment, the barcode **222** is used to verify that the item **12** is an item of the order. In an embodiment, the label **45** is verified. In an embodiment, the images of the item **12** used for verification are compiled in a file and sent to a pharmacist who compares the images with the order information and uses an input device to confirm or approve of the label item **12** or reject or except the labeled item **12**, which sends the item to be reviewed. This verification may occur at any time after the labeled item **12** exits the label applicator assembly **40** and prior to being sent to the patient or customer.

[0122] FIG. **14c** shows an embodiment of a packaging process **860** that includes verifying the item **12**, and packaging one or more items **12** of an order in a bag or package.

[0123] At step **862**, each item **12** arrives at the verification station **500** where the label **45** and on each item **12** is imaged for verification.

[0124] For the packaging steps, the package **615** is prepared for each order. At step **864**, the bagging assembly **600** prints order identification information onto the package **615** and reads and verifies that the order identification information matches the order to be advanced into the package **615**. In an embodiment, a package verification imaging device **660** is communicatively coupled to the processing system **15** and verifies the package.

[0125] At step **866**, the packaging assembly **50** opens the verified package **615**.

[0126] At step **868**, the item **12** indexes or advances onto the gate **555** that is in covering

[0127] relationship with the opening or drop gate zone **149**, the gate **555** is opened, and the item **12** is dropped through the chute **505** and into the bag or package **615**.

[0128] In an embodiment, at step **870** the chute imaging device **620** detects if an item is stuck in the chute **505** or verifies that the chute **505** is free of jams or obstructions.

[0129] For the document printing steps, the one or more patient information documents are prepared for each order. At step **872**, the document printing station prints the patient information documents, including unique document identification indicia on each page of the documents for the order, reads and verifies that the document identification indicia on the documents matches the order, and transfers or advances the documents into the package. In an embodiment, a document verification imaging device is communicatively coupled to the processing system **15** and verifies the patient information documents.

[0130] At step **874**, the packaging assembly **50** seals the package **615** and drops the package **615** with the order identification information facing upward onto a bidirectional conveyor. The package verification imaging device **660** identifies whether the package containing the one or more items of the order is verified in step **876** and transfers verified packages **615** onto a conveyor to exit the IPM

assembly **10** in step **878** or identifies and transfers the one or more items that are unverifiable or previously rejected to a review conveyor for review at step **880**.

[0131] Many different arrangements of the various components depicted, as well as components not shown, are possible without departing from the spirit and scope of the present disclosure. Embodiments of the present disclosure have been described with the intent to be illustrative rather than restrictive. Alternative embodiments will become apparent to those skilled in the art that do not depart from its scope. A skilled artisan may develop alternative means of implementing the aforementioned improvements without departing from the scope of the present disclosure. It will be understood that certain features and subcombinations are of utility and may be employed without reference to other features and subcombinations and are contemplated within the scope of the claims.

Claims

1. An item packaging machine configured to fulfill a plurality of orders, wherein each of the plurality of orders includes one or more items each having an item identifying indicia thereon, the item packaging machine comprising: a processing system accessing order information associated with each of the plurality of orders and including order identifying indicia for each of the plurality of orders; a conveyor configured to advance the one or more items through the item packaging machine, wherein the conveyor forms a conveyor run on which each of the one or more items are advanced; an introduction assembly configured to associate an indexed position of each of the one or more items positioned on the conveyor run with item identifying information for the item at the respective indexed position; a label applicator positioned proximate a labeling station on the conveyor run; at least one labeling verification camera positioned proximate the label applicator and in communication with the processing system, wherein upon each of the items being advanced to the labeling station, the processing system causes the at least one labeling verification camera to capture the item identifying indicia on the item at the labeling station and if the item identifying indicia captured from the item at the labeling station corresponds to the item identifying indicia associated in the processing system with the item expected at the labeling station based upon the indexed position of the item in the processing system, the processing system causes the label applicator to apply to the item a label with customer identifying indicia associated with the item printed thereon and thereafter the processing system causes the at least one labeling verification camera to capture at least one image of the item wherein the at least one image includes the item identifying indicia and at least a portion of the label including the customer identifying indicia thereon; and a packaging verification camera positioned proximate a packaging verification station on the conveyor run proximate a packaging assembly operable to advance the item from the packaging verification station into a package, the packaging verification camera and the packaging assembly in communication with the processing system, the processing system causes the packaging verification camera to capture the item identifying indicia on the item or the customer identifying indicia on the label at the packaging verification station and if the item identifying indicia or the customer identifying indicia captured from the item at the packaging verification station corresponds to the item identifying indicia associated in the processing system with the item expected at the packaging verification station based upon the indexed position of the item in the processing system, the processing system causes the packaging assembly to advance the item into a package and to apply the order identifying information for the order to the package.
2. The item packaging machine of claim 1, wherein the conveyor further comprises a plurality of V-shaped blocks and the indexed position of the item corresponds to one of the plurality of V-shaped blocks.
3. The item packaging machine of claim 1, wherein the introduction assembly comprises an introduction camera that is communicatively coupled to the processing system, wherein the

processing system causes the introduction camera to capture an image of the item identifying indicia on each of the items as the item is advanced into one of the indexed positions on the conveyor run to associate the item identifying indicia with the indexed position and to verify the item identifying indicia corresponds to the item identifying indicia of one of the one or more items in the order.

4. The item packaging machine of claim 3, further comprising a display communicatively coupled to the processing system, wherein upon capturing the image of the item identifying indicia of the item by the introduction camera, the display provides a prompt indicating an orientation of the item identifying indicia on the item relative to the conveyor when positioned on the conveyor run.

5. The item packaging machine of claim 1, wherein the at least one labeling verification camera comprises a set of six labeling verification cameras wherein each of the six labeling verification cameras is positioned relative to the labeling station to image a different side of the item.

6. The item packaging machine of claim 1, wherein the label applicator is mounted for movement along at least two axes, two axes extending in a plane extending generally perpendicular to the conveyor run across the labeling station.

7. The item packaging machine as in claim 1 wherein the conveyor run across the labeling station comprises an item receiving platform advanceable along an axis perpendicular to the conveyor run across the labeling station for raising or lowering the item receiving platform relative to the conveyor run at the labeling station; and the at least one labeling verification camera comprises a set of six labeling verification cameras wherein each of the six labeling verification cameras is positioned relative to the labeling station to image a different side of the item on the item receiving platform.

8. The item packaging machine of claim 7, wherein the order information accessed by the processing system includes a print file including the customer identifying indicia and patient-specific information for each of the one or more items in the order to be printed on the label, a location for applying the label containing the patient-specific information and the customer identifying indicia relative to a side of the item positioned on the item receiving platform, and a position of the item receiving platform relative to the conveyor run to position the item so that one of the six labeling verification cameras can capture an image the side of the item on which the item identifying information is located.

9. The item packaging machine of claim 1, wherein the packaging assembly further comprises a document verification camera positioned proximate the packaging assembly and in communication with the processing system, the processing system causing the document verification camera to capture an image of a document verification indicia applied to one or more patient information documents associated with at least one of the items in the order and to compare the document verification indicia captured by the document verification camera to the document verification indicia associated with each of one or more patient information documents associated with the order in the processing system prior to insertion of the one or more patient information documents into the package for each order.

10. A method of fulfilling a plurality of orders that each include one or more items for processing in an item packaging machine, the method comprising: retrieving from a database, order information regarding an order including item identification information corresponding to each of the one or more items contained in the order and customer information of a customer associated with the order; imaging an item identification indicia of each of the one or more items in succession, wherein each item identification indicia is associated with the item identification information for a respective one of the items in the order; for each of the one or more items whose item identification indicia is imaged, prompting placement of the item onto a conveyor in a specified orientation; positioning the item of the order onto the conveyor with the item identification indicia on the item in the specified orientation and at a specified position on the conveyor; conveying the item to a verification position on the conveyor; at the verification position, imaging the item identification

indicia on the item and comparing the item identification indicia to the item identification information in the order information retrieved from the database and if the item identification indicia on the item corresponds to the item identification information in the order, applying customer information associated with the item of the order to the item; and after applying the customer information to the item and with the item at the verification position, capturing verification images that include an image of the item identification indicia on the item and an image of the customer information applied to the item.

11. The method of claim 10, wherein the item identification indicia and the customer information are imaged simultaneously.

12. The method of claim 10, wherein the verification position is on a vertically moving platform.

13. The method of claim 10, further comprising performing a pharmaceutical verification of the item, the pharmaceutical verification including: imaging a selected side of the item prior to applying the customer information to the item; compiling into a file the verification images, the image of the selected side of the item prior to applying the customer information thereto, and a pre-existing image from the item identification information from the database that corresponds to the selected side of the item; comparing the images in the file with at least the item identification information retrieved from the database; and confirming the item is the item in the order or sending the item to be reviewed.

14. The method of claim 10, wherein the order information retrieved from the database includes address information for the customer and the method further comprises: conveying the item to a packaging verification position; at the packaging verification position, imaging the customer information on the item and comparing the customer information to the item identification information in the order information retrieved from the database and if the customer information on the item at the packaging verification position corresponds to the item identification information in the order for the item expected at the packaging verification position, advancing the item into a package; and applying the address information for the customer on the package into which the item is placed.

15. The method as in claim 14 wherein the order information retrieved from the database includes order identification indicia and the method further comprises applying the order identification indicia on the package into which the item is to be placed and imaging the order identification indicia.

16. A method of fulfilling a plurality of orders that each include one or more items for processing in an item packaging machine, the method comprising: receiving an order container at an item introduction assembly containing one or more items corresponding to an order and imaging a container identification indicia of the order container; retrieving from a database order information associated with the container identification indicia imaged at the item introduction assembly, the order information corresponding to the order comprising the one or more items contained in the order container imaged at the item introduction assembly; imaging an item identification indicia of each of the one or more items retrieved in succession from the order container; for each of the one or more items whose item identification indicia is imaged, prompting placement of the item onto a conveyor in a specified orientation; placing the item onto the conveyor in a specified position in the specified orientation; verifying a presence of the item on the conveyor; indexing the specified position of the item on the conveyor; incrementally advancing the conveyor with the item in the specified position towards a labeling position; upon advancement of the item to the labeling position, verifying the item and the specified orientation of the item by imaging the item identification indicia on the item and comparing the item identification indicia to the order information and to the item expected in the specified position; applying patient-specific information to the item in the labeling position; verifying the item by imaging, in the labeling position, the patient-specific information applied to the item and comparing the patient-specific information to the order information; incrementally advancing the conveyor with the item in the

specified position towards a packaging assembly; at the packaging assembly, verifying the item by imaging the patient-specific information applied to the item and comparing the patient-specific information to the order information and to the item expected in the specified position; applying order identification information to a package; verifying the package by imaging the order identification information applied to the package and comparing the order identification information with the order information; advancing the one or more items of the order into the package; and sealing the package.

17. The method of claim 16, wherein the conveyor includes a plurality of V-shaped blocks, and the specified position of the item corresponds to one of the plurality of V-shaped blocks.

18. The method of claim 16, further comprising verifying the item by imaging the item identification indicia of the item simultaneously with imaging of the patient-specific information applied to the item and comparing the item identification indicia to the order information.

19. The method of claim 18, wherein verifying the item further comprises imaging at least one surface of the item prior to applying the patient-specific information thereto; compiling a set of images that includes: the image of the patient-specific information applied to the item, the image of the item identification indicia of the item taken simultaneously with the image of the patient-specific information applied to the item, the image of the at least one surface of the item prior to applying the patient-specific information thereto, and at least one pre-existing image that corresponds to the image of the at least one surface of item; comparing the set of images with the order information; and operating an input device to confirm the labeled item is the item in the order or send the item for review.

20. The method of claim 16, wherein the labeling position is on an item receiving platform that is vertically positionable in an upper position and a lowered position, and imaging in the labeling position is effected by a set of one or more cameras that are vertically positionable with the item receiving platform.

21. The method of claim 16, further comprising printing one or more patient information documents with one or more printers, imaging each of the one or more patient information documents to verify that the one or more patient information documents is associated with the one or more items of the order, and advancing the one or more patient information documents into the package prior to sealing the package.
